Fu-Te Wong

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Profile

AI/ML engineer and researcher (10+ yrs) specialising in multi-agent self-improving LLM systems (alignment and interpretability), quantum-integrated ML, and deep RL. PhD candidate at University of Toronto (Vector Institute affiliate). Shipped end-to-end ML pipelines on cloud infra; experienced in cluster orchestration, MLOps, and infrastructure. Proven track record delivering AI research, building end-to-end ML pipelines.

Technical Skills

ML / AI	PyTorch, JAX, TensorFlow-Quantum, Scikit-learn, Hugging Face, LLM fine-tuning & agents, Multi-agent RL, LangChain, Qiskit, PennyLane
Stack	Python, C/C++, Java, SQL, JavaScript/TypeScript, React, Next.js, Django REST, Bash/Shell
CI/CD	Git, GitHub Actions, Docker, Docker Compose; experience with CI pipelines for ML model validation and automated test suites
Infra / MLOps	GCP, AWS (S3/EC2), Kubernetes, Hadoop/Spark (CDH), TensorflowOnSpark, Linux sysadmin, cluster provisioning & job scheduling, vLLM, SGLang, megatron, verl
Data	NumPy, Pandas, SciPy; multimodal neuroimaging (fMRI/MRI/MEG/DTI); topological & dynamical systems analysis

Experience

Researcher *2024 – Present*

Vector Institute for Artificial Intelligence, Toronto

- Built multi-agent LLM/RL pipeline for automated QML algorithm search; abstracted a novel *advanced for-loop* construct accepted at QTML 2025 (arXiv:2506.18260)
- Designed agentic search framework with modular tool-use, prompt chaining, and self-evaluation loops; architect inference system; authored quantum backpropagation formulation (under review)
- Formulating artificial consciousness via Integrated Information Theory
- Co-organiser, AI-for-Reasoning reading group; driving LLM multi-token prediction on multi-hop reasoning and suicidality-detection agent projects; DPO variant (WIP)

Ph.D. Researcher — CS & Medical Science *2021 – 2026*

University of Toronto (thesis: Agentic system for exploration of algorithmic ANNs)

- Contributed to generative quantum eigensolver (GQE) (arXiv:2401.09253)
- Engineered virtual brain stimulation AI; integrated E-field modelling with connectome tractography for TMS target selection

Teaching Assistant — Deep Learning *2020 – 2023*

Neuromatch Academy (Lead TA 2021; global cohort of 1,000+ students)

- Delivered DL curriculum (PyTorch); led debugging sessions and model evaluation workshops across distributed online cohorts

Data Analyst *2019 – 2021*

Institute of Linguistics, Academia Sinica, Taiwan

- Designed and deployed multimodal neuroimaging pipelines (MRI/fMRI/DTI/MEG) for cross-age working-memory study; reduced manual QC time by ~40% through automated topological analysis scripts

Research Assistant — Medical AI *2018-2019*

Center for AI in Medicine, Taipei Medical University

- Delivered CNN-based diagnostic model for idiopathic normal-pressure hydrocephalus and brain-tumour segmentation; published CNN suicidality-prediction model (*J. Clinical Psychiatry*, 82(2), 2021)
- Built and documented end-to-end data pipeline: DICOM ingestion → preprocessing → model inference → clinical reporting

Software Engineer *2017*

Cybersecurity Technology Institute, Institute for Information Industry, Taiwan

- Developed security tooling and data-processing scripts in Python/C++ within an agile team; participated in code review and issue tracking workflows

Visiting Student — Big Data & ML Infrastructure *2016 – 2017*

Modeling & Informatics Lab, National Taiwan University

- Provisioned and administrated Hadoop/Spark CDH cluster for large-scale neural image analysis (Thunder-project, TensorflowOnSpark); aligned chatbot LU/NLG embeddings with brain activity signals

Clinical Psychologist *2011 – 2013*

Taipei Veterans General Hospital

- Psychiatry Department, Neurological Institute, and Rehabilitation Center

Selected Projects

MetaScientist (Meta_Scientist): Self-improving agentic research framework pairing LLM multi-agent orchestration with Monte-Carlo tree search (MCTS) to automatically search quantum machine-learning algorithm pipelines; integrates web search, task planning, and a dual-loop research cycle. Implementation framework behind the QTML 2025 paper; extends MetaGPT/SELA. *Tech: Python, MetaGPT, MCTS, Docker*

DDB Knowledge Agent (ddb-knowledge-agent-py): Streaming RAG agent fusing pgvector similarity retrieval with Postgres full-text search via reciprocal rank fusion (RRF); ReAct tool calling, query-complexity-tiered model routing, semantic caching, and cost auditing; PostgreSQL runs under rootless Apptainer, deployable on Docker-less HPC clusters. *Tech: Python, FastAPI, SSE, Gemini, PostgreSQL/pgvector, Apptainer*

GOAT Network Agent (goat-agent_demo): Autonomous Web3 agent registering an ERC-8004 on-chain identity and wired into the x402 payment flow; executes price tracking → conditional trigger → on-chain payment → attested settlement end-to-end with no human intervention after launch. *Tech: Node.js, Express, ERC-8004, x402*

Education

Ph.D. Candidate, Institute of Medical Science, University of Toronto, Canada 2021 – 2026
M.S., Clinical Psychology, National Chung Cheng University, Taiwan 2009 – 2012

Selected Publications

Wong, F.-T. (2026). Quantum Backpropagation. *(Under review).*
Frydenlund, A., Wong, F.-T., et al. (2026). Multi-Token Prediction and Alternative Tokenization Improve Multi-Hop Reasoning and Graph Search. *(Under review).*
Wong, F.-T. (2025). Advanced For-Loop for QML algorithm search. *arXiv:2506.18260.* QTML 2025.
Nakaji, K., …, Wong, F.-T., et al. (2024). GQE. *arXiv:2401.09253.*
Chen, V.C.-H., Wong, F.-T., et al. (2021). CNN suicidality prediction. *J. Clin. Psychiatry* 82(2).

Awards

Vector Institute Research Scholarship (2024, 2025)
Pixnet Hackathon: A.I. Image Gen. (2018) - 2nd Prize
Pixnet Hackathon: A.I. Cloze (2016) - 1st Prize